Optimization of Single-Variable Functions

SECON 320 - Introduction to Mathematical Economics

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Contents

1 Optimum Values and Extreme Values

An economic agent chooses a scalar decision variable x (output Q, advertising A, savings s, etc.) to maximize or minimize an objective function $y = f(x)$. The optimization target may be a *relative* (local) extremum or an *absolute* extremum over the domain.

Definition 1 (Relative (Local) vs. Absolute Extremum)**.** Let $f : \mathcal{D} \subseteq \mathbb{R} \to \mathbb{R}$ and $x_0 \in \mathcal{D}$.

- $f(x_0)$ is a *relative (local) maximum* if $\exists \varepsilon > 0$ s.t. $f(x_0) \geq f(x)$ for all x with $|x - x_0| < \varepsilon$.
- $f(x_0)$ is a *relative (local) minimum* if $\exists \varepsilon > 0$ s.t. $f(x_0) \leq f(x)$ for all x with $|x - x_0| < \varepsilon$.
- $f(x^\star)$ is an *absolute maximum (minimum)* if $f(x^\star) \geq (\leq)\, f(x)$ for all $x \in \mathcal{D}$.

Economist's Intuition 1. In many economic problems the domain is restricted (e.g., $Q \geq 0$). Absolute optima may then occur at *boundaries* (corner solutions). Always check feasible endpoints.

2 Relative Maximum and Minimum: First-Derivative Test

Stationary points and the sign-change test

If f is differentiable, any interior relative extremum must occur at a *stationary point* x_0 where

$$f'(x_0) = 0.$$

This is *necessary* but not sufficient: $f'(x_0) = 0$ could also be an inflection point. The first-derivative *sign-change* criterion turns necessity into a decision rule.

Proposition 1 (First-Derivative Test)**.** Let f be differentiable around x_0 with $f'(x_0) = 0$.

- If f' changes $+$ to $-$ when crossing x_0, then x_0 is a relative maximum.
- If f' changes $-$ to $+$ when crossing x_0, then x_0 is a relative minimum.
- If f' does not change sign across x_0, then x_0 is neither max nor min (typically an inflection).

Example 1 (Cubic with two stationary points)**.** Let $f(x) = x^3 - 12x^2 + 36x + 8$. Then $f'(x) = 3x^2 - 24x + 36 = 3(x-2)(x-6)$, so the stationary points are $x = 2, 6$. A sign chart for f' shows: on $(-\infty, 2)$, $f' > 0$; on $(2, 6)$, $f' < 0$; on $(6, \infty)$, $f' > 0$. Hence

$$x = 2 \text{ is a local maximum,} \qquad x = 6 \text{ is a local minimum.}$$

Evaluating: $f(2) = 2^3 - 12(4) + 72 + 8 = 8 - 48 + 72 + 8 = 40$, $f(6) = 216 - 432 + 216 + 8 = 8$.

Example 2 (Average cost minimization)**.** Average cost $AC(Q) = Q^2 - 5Q + 8$. Then $AC'(Q) = 2Q - 5$, so the unique stationary point is $Q^* = 2.5$. Check the sign change of AC': negative for $Q < 2.5$, positive for $Q > 2.5 \Rightarrow Q^*$ minimizes AC and, for this quadratic, is also the absolute minimum.

3 Second and Higher Derivatives

The second derivative measures the rate of change of the slope:

$$f''(x) = \frac{\mathrm{d}}{\mathrm{d}x}[f'(x)], \qquad f^{(n)}(x) = \frac{\mathrm{d}[}{\mathrm{d}n}]f(x)x.$$

Geometry and curvature

- $f''(x) > 0 \Rightarrow$ slope increasing: the graph is *convex* (strictly convex if $f'' > 0$ on an interval).
- $f''(x) < 0 \Rightarrow$ slope decreasing: the graph is *concave* (strictly concave if $f'' < 0$ on an interval).
- At an *inflection point* the curvature changes sign; f' typically reaches a local extremum.

Economist's Intuition 2 (Risk attitudes)**.** In expected-utility models, a concave utility U ($U'' < 0$) captures risk aversion; a convex utility ($U'' > 0$) captures risk loving. For a fair gamble with expected payoff $E[x]$, a risk-averse agent prefers the certain $E[x]$ to the lottery because diminishing marginal utility makes the utility of the gain smaller in absolute value than the utility loss of an equally likely loss.

Example 3 (Interpreting f' and f'' numerically)**.** If $f'(x_0) = 1.2$ and $f''(x_0) = -0.5$, then near x_0 the function is increasing ($f' > 0$) but at a diminishing rate ($f'' < 0$): think increasing output with diminishing marginal returns.

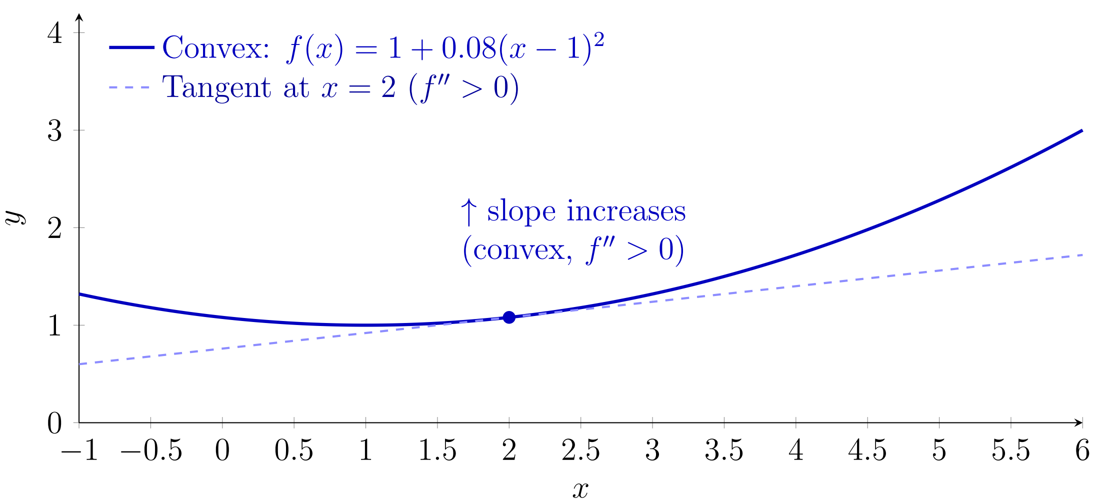


Figure 1: Convexity ($f'' > 0$): the slope increases with x. The dashed line is the tangent at $x = 2$.

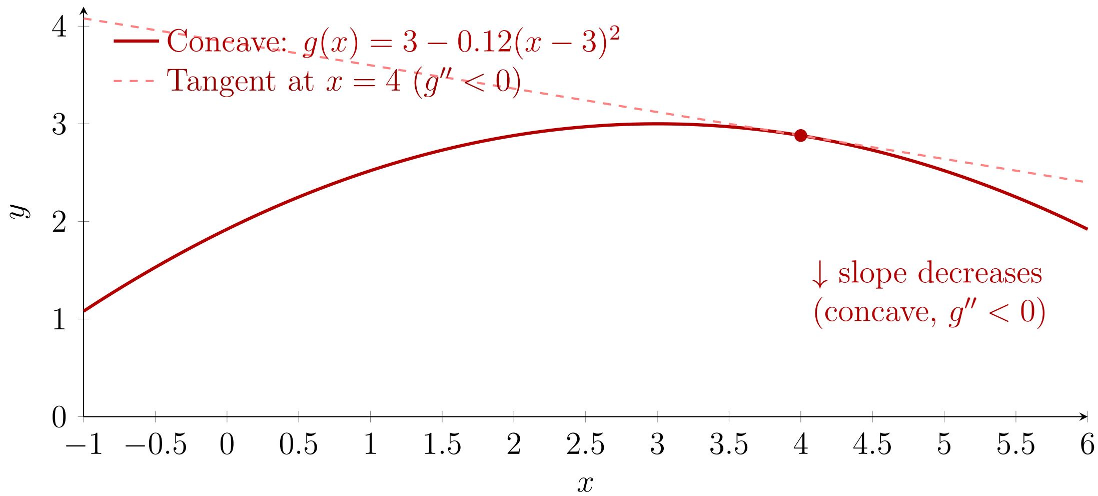
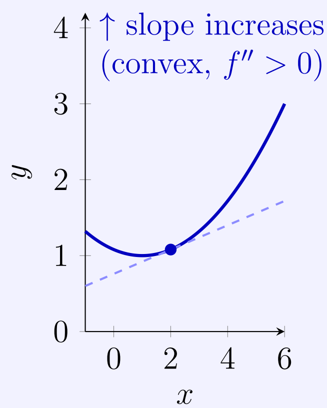


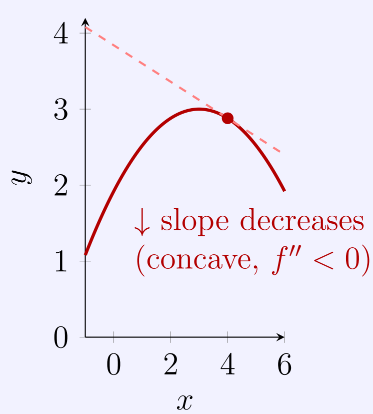
Figure 2: Concavity ($g'' < 0$): the slope decreases with x. The dashed line is the tangent at $x = 4$.

Convexity ($f'' > 0$)



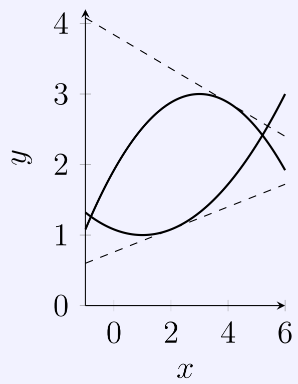
If $f'(x_0) = 0$ and $f''(x_0) > 0$, x_0 is a *local minimum* (Chs. 9.3–9.4). When $f''(x_0) = 0$ is inconclusive, use the sign change of f' (Ch. 9.2) or the nth-derivative test (Ch. 9.6).

Concavity ($f'' < 0$)



If $f'(x_0) = 0$ and $f''(x_0) < 0$, x_0 is a *local maximum*. Geometrically: the slope is decreasing; f' switches from $+$ to $-$ across the peak (Ch. 9.2–9.4).

Curvature & Tangents



If $f'(x_0) = 0$ and $f''(x_0) > 0$ (convex), x_0 is a local minimum; if $f''(x_0) < 0$ (concave), it's a local maximum. Use the sign-change of f' when $f''(x_0) = 0$.

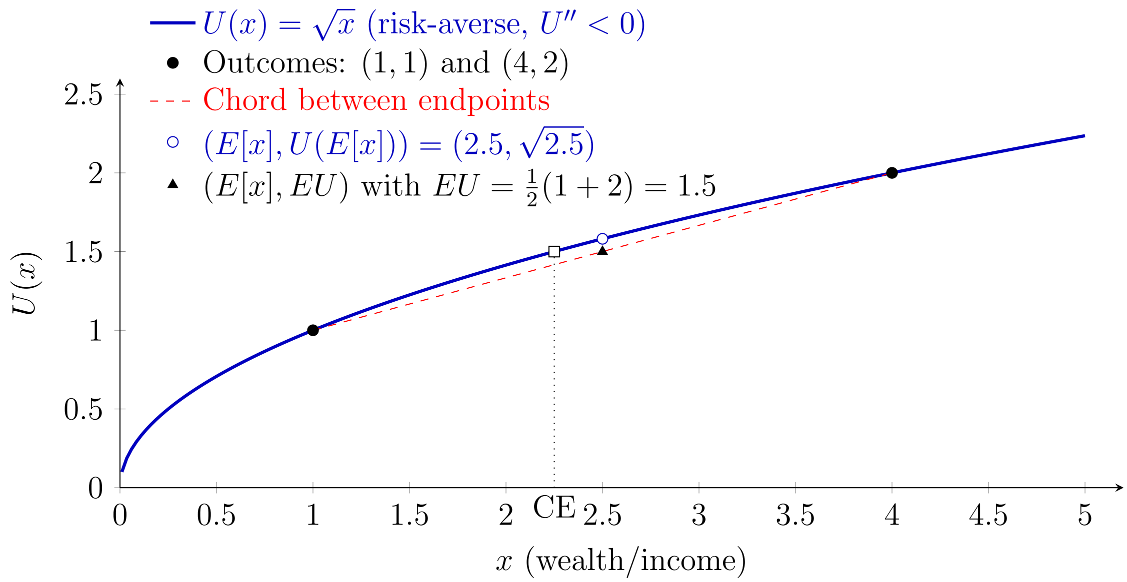
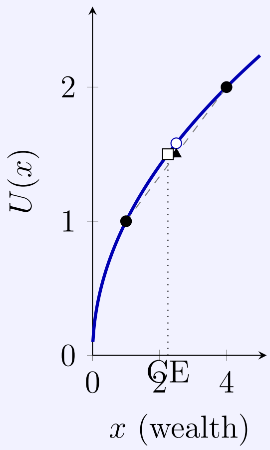


Figure 3: Risk Aversion: For a 50–50 gamble between $x = 1$ and $x = 4$, $EU = 1.5 < U(E[x]) = \sqrt{2.5}$. The certainty equivalent (CE = 2.25) is less than $E[x] = 2.5$.

Risk Aversion & CE



Concavity ($U'' < 0$) $\Rightarrow$ diminishing marginal utility (Ch. 9.3).
Jensen: $U(E[x]) \geq E[U(x)]$ so the certainty equivalent CE solving $U(\text{CE}) = EU$ satisfies CE $< E[x]$.
The vertical gap to the chord is the *risk premium*.

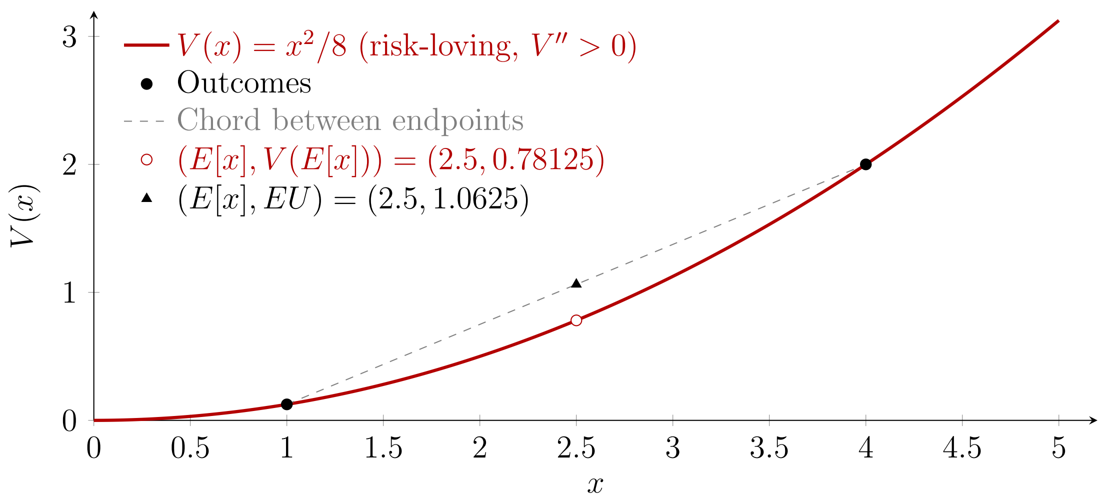
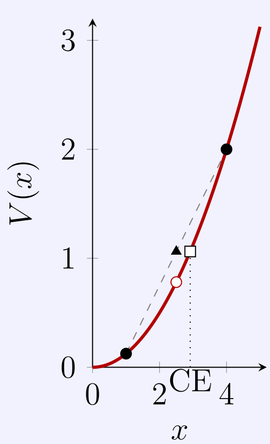


Figure 4: Risk Loving: $EU > V(E[x])$, reflecting preference for risk when $V'' > 0$.

Risk Loving (contrast)



With convex V ($V'' > 0$), the inequality reverses: $V(E[x]) \leq E[V(x)]$ (Ch. 9.3). Risk lovers have CE $> E[x]$. Curvature alone flips preferences over lotteries vs. their mean.

4 Second-Derivative Test; Necessary vs. Sufficient Conditions; Profit Maximization

Second-derivative test

If f is twice differentiable and $f'(x_0) = 0$:

$$\begin{cases} f''(x_0) < 0 \Rightarrow x_0 \text{ is a relative maximum,} \\ f''(x_0) > 0 \Rightarrow x_0 \text{ is a relative minimum,} \\ f''(x_0) = 0 \Rightarrow \text{inconclusive (use higher derivatives or the sign-change test).} \end{cases}$$

Necessary vs. sufficient

- $f'(x_0) = 0$ is a *necessary* interior condition (first-order condition, FOC).
- Together with $f''(x_0) \lesseqgtr 0$, we obtain a *sufficient* condition (second-order condition, SOC) for a local max/min.

Application: profit maximization and the MR=MC rule

Let total revenue $R(Q)$ and total cost $C(Q)$ be differentiable; profit $\pi(Q) = R(Q) - C(Q)$. The FOC

$$\frac{\partial \pi}{\partial Q} = R'(Q) - C'(Q) = 0 \quad \Longleftrightarrow \quad MR = MC$$

pins down *candidates*. The SOC for a maximum is $\pi''(Q) = R''(Q) - C''(Q) < 0$ at the candidate.

Example 4 (Profit with a cubic cost)**.** Suppose $R(Q) = 1200Q - 2Q^2$ and $C(Q) = Q^3 - 61.25Q^2 + 1528.5Q + 2000$. Then

$$\pi'(Q) = R'(Q) - C'(Q) = (1200 - 4Q) - (3Q^2 - 122.5Q + 1528.5) = -3Q^2 + 118.5Q - 328.5.$$

Setting $\pi'(Q) = 0$ gives two stationary outputs (use quadratic formula). The SOC

$$\pi''(Q) = R''(Q) - C''(Q) = -4 - (6Q - 122.5) = -6Q + 118.5$$

selects the one with $\pi''(Q) < 0$ as the profit maximum. Always confirm the economic domain $(Q \geq 0)$ and compare profit values if boundaries matter.

Economist's Intuition 3 (Upward-sloping MR can occur)**.** When $AR(Q)$ is strictly convex on a segment, $MR(Q) = AR(Q) + Q \cdot AR'(Q)$ can slope upward on that segment because the $Q\,AR'(Q)$ term can dominate; nonetheless, MR=MC must still hold at an interior maximum, along with $\pi'' < 0$.

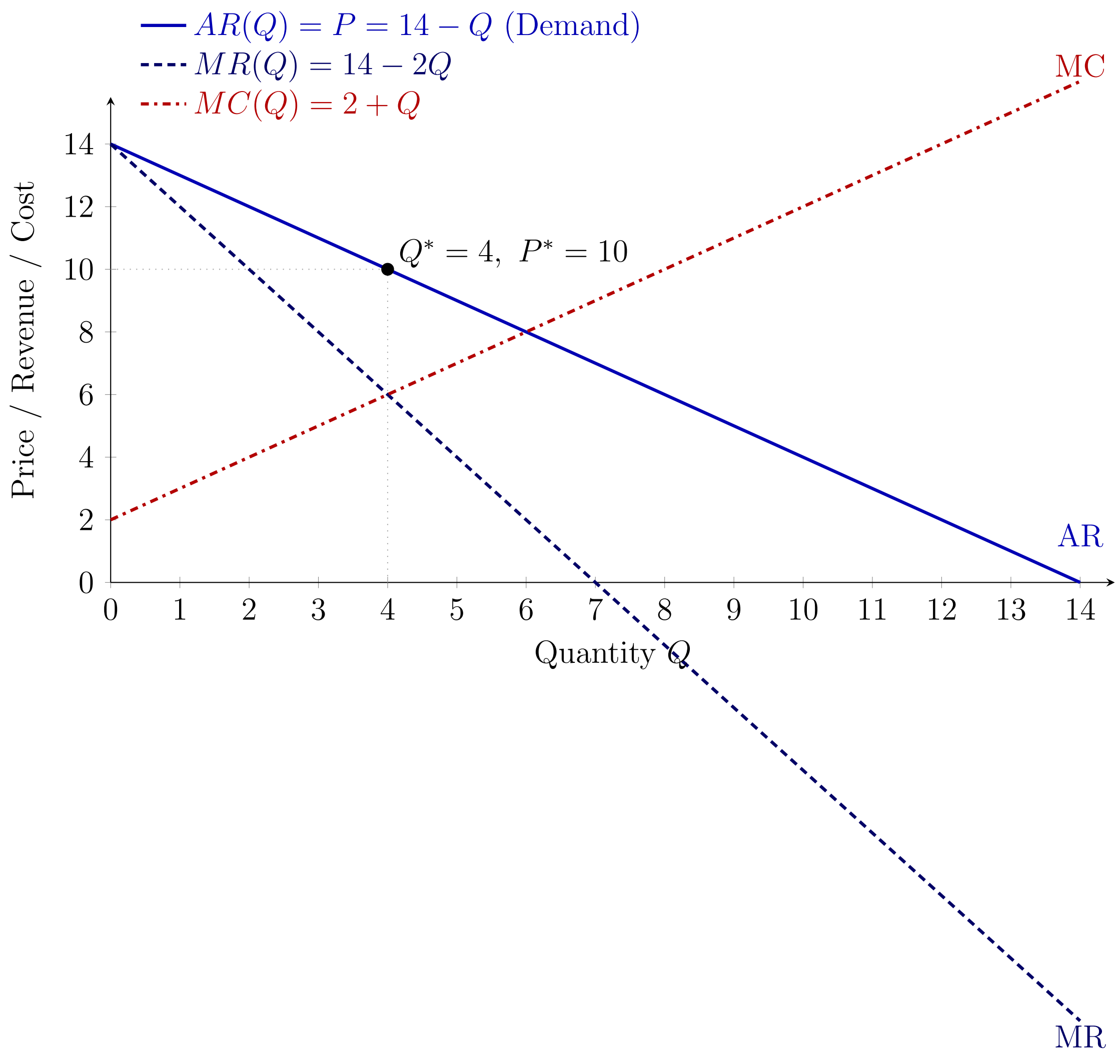
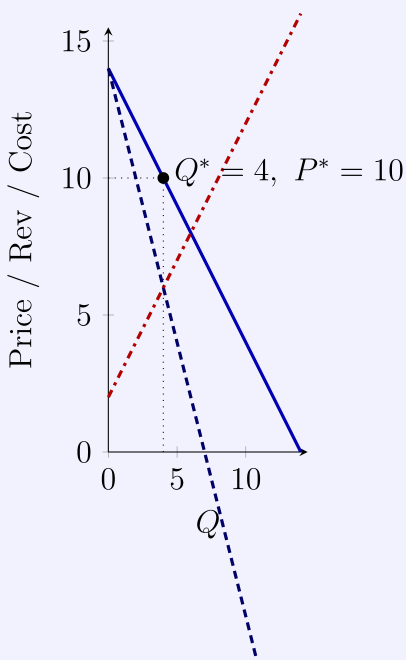


Figure 5: Monopoly choice: Interior optimum where $MR(Q^*) = MC(Q^*)$ with P^* from the demand curve.

MR–MC Geometry



FOC (Ch. 9.4): $MR = MC$ selects candidates. SOC: $\pi''(Q) = R''(Q) - C''(Q)$. Here $R(Q) = 14Q - Q^2 \Rightarrow R'' = -2$; with $MC = 2 + Q$ we get $C'' = 1$, so $\pi'' = -3 < 0$—a strict local maximum. Always compare with boundaries (capacity, $Q \geq 0$) for absolute optimality.

5 Maclaurin and Taylor Series (Approximations and Remainders)

Motivation

When $f''(x_0) = 0$ (or higher derivatives vanish), the standard second-derivative test is inconclusive. Taylor expansions provide a principled way to examine higher-order curvature and to build local polynomial approximations.

Factorials and notation

For $n \in \mathbb{N}$, $n! = n(n-1)\cdots 2 \cdot 1$ with $0! \equiv 1$. We write $f^{(k)}$ for the k-th derivative.

Maclaurin expansion (around 0)

If f is infinitely differentiable at 0, its Maclaurin polynomial of order n is

$$P_n(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n.$$

Example. If $f(x) = 2 + 4x + 3x^2$, then $f(0) = 2$, $f'(0) = 4$, $f''(0) = 6$, so its Maclaurin series reproduces f exactly.

Taylor expansion (around x_0)

More generally,

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + R_n(x),$$

where the *Lagrange remainder* is

$$R_n(x) = \frac{f^{(n+1)}(p)}{(n+1)!}(x - x_0)^{n+1} \quad \text{for some } p \text{ between } x \text{ and } x_0.$$

Example 5 (Non-polynomial approximation)**.** Expand $\phi(x) = \dfrac{1}{1+x}$ around $x_0 = 1$ up to order $n = 4$. Compute derivatives $\phi^{(k)}(x) = (-1)^k k!(1+x)^{-(k+1)}$. Evaluate at $x_0 = 1$ and substitute into Taylor's formula to obtain a quartic approximation plus $R_4(x)$. This gives an excellent local approximation near $x_0 = 1$ and quantifies the error via R_4.

Economist's Intuition 4 (Linearization you will use a lot)**.** For small Δ, $f(x_0 + \Delta) \approx f(x_0) + f'(x_0)\Delta$. This is the *tangent-line* (first-order) approximation used constantly in comparative statics and elasticities.

6 The nth-Derivative Test for Relative Extremum of One Variable

When lower-order derivative tests fail (e.g., $f'(x_0) = 0$ and $f''(x_0) = 0$), Taylor's theorem yields a powerful generalization.

Proposition 2 (General nth-derivative test)**.** Let f be n times continuously differentiable near x_0 with

$$f'(x_0) = f''(x_0) = \cdots = f^{(n-1)}(x_0) = 0, \qquad f^{(n)}(x_0) \neq 0.$$

- If n is *even* and $f^{(n)}(x_0) > 0$, then x_0 is a relative *minimum.*
- If n is *even* and $f^{(n)}(x_0) < 0$, then x_0 is a relative *maximum.*
- If n is *odd*, then x_0 is an *inflection point* (no extremum).

Example 6 (Degenerate second derivative)**.** Consider $f(x) = x^4$. Then $f'(0) = 0$, $f''(0) = 0$, $f^{(3)}(0) = 0$, $f^{(4)}(0) = 24 > 0$. The first nonzero derivative is fourth and positive $\Rightarrow x = 0$ is a (flat) local minimum.

Boundary and corner solutions (one dimension)

If the feasible set is a closed interval $[a, b]$, compare interior stationary values with $f(a)$ and $f(b)$. If the optimum occurs at a or b, the FOC $f' = 0$ need not hold: the Kuhn–Tucker logic in one dimension reduces to checking the boundary.

A checklist for single-variable optimization

1. Specify objective $f(x)$ and feasible domain $\mathcal{D}$ (include units and economic meaning).
2. Compute $f'(x)$, solve $f'(x) = 0$ for interior candidates; include any points where f' is undefined but $x \in \mathcal{D}$.
3. Classify each candidate: use the second-derivative test; if inconclusive, use the first-derivative sign change or the nth-derivative test.
4. Evaluate f at boundaries and compare with interior values for absolute optima.
5. In applications (profit, cost, utility), translate conditions back: MR=MC and $\pi'' < 0$ for a maximum; AC minimization at Q where $AC' = 0$ and $AC'' > 0$, etc.

Exercises

1. Find and classify extrema of $f(x) = x^3 - 3x + 1$. *Answer:* $f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$. $f''(x) = 6x$: at $x = -1$, $f'' < 0$ (local max); at $x = 1$, $f'' > 0$ (local min).

2. Let $AC(Q) = Q - 4 \ln Q + 7$ for $Q > 0$. Minimize AC. *Answer:* $AC'(Q) = 1 - 4/Q = 0 \Rightarrow Q^* = 4$. $AC''(Q) = 4/Q^2 > 0$, so Q^* is a strict minimum.

3. (Risk) For $U(x) = \sqrt{x}$ and a fair 50–50 bet between 10 and 20, compare EU with $U(E[x])$. *Answer:* $EU = \frac{1}{2}(\sqrt{10} + \sqrt{20}) < \sqrt{15} = U(15)$ (risk-averse).

4. (Degenerate curvature) Classify $f(x) = x^6 - 3x^4$. *Answer:* $f'(x) = 6x^5 - 12x^3 = 6x^3(x^2 - 2)$ gives $x \in \{0, \pm\sqrt{2}\}$. $f''(x) = 30x^4 - 36x^2$: at $x = 0$, $f'' = 0$ (use higher order: $f^{(4)}(0) = -72 < 0 \Rightarrow$ local maximum at 0). At $\pm\sqrt{2}$, $f'' > 0 \Rightarrow$ local minima.

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